

Counterfactual Explainability: Deep RL vs. Online Planning on MiniGrid

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Abstract—This paper evaluates the relative explainability of online planning and deep reinforcement learning when both agents are asked to justify the same decision under an identical Markov decision process. A proximal-policy-optimisation (PPO) agent and a from-scratch Monte Carlo Tree Search (MCTS) planner are operated on `MiniGrid-Dynamic-Obstacles-8x8`, both trained with the same discount factor $\gamma = 0.99$ on the same per-step-penalised reward so the two solvers optimize the identical objective. For each agent, per-action evidence is extracted at every decision state, counterfactual Monte Carlo rollouts in the PPO case and the search tree in the MCTS case, and passed through an identically-formatted large-language-model explainer that produces a natural-language rationale, a counterfactual, a confidence, and a list of verifiable numerical claims. MCTS evidence additionally exposes the top-3 principal variations from the root, the depth-2 visit distribution, and per-child return variance. Explanations are scored by an independent LLM judge on three metrics: fidelity, soundness, and a post-hoc inferability test stricter than the prior version (verbal action names are now also scrubbed). Across 290 decision states drawn from two *performance-matched* agents (PPO 1.000 task-success, MCTS 1.000 task-success, both at +0.941 mean unshaped return and 16.8 mean steps), the two evidence sources produce *different* explanation quality on every metric: PPO-rollout evidence yields fidelity 0.854 [95% CI 0.83, 0.88], soundness 0.852 [0.83, 0.88], inferability 0.837 [0.77, 0.89]; MCTS-tree evidence yields fidelity 0.787 [0.76, 0.82], soundness 0.895 [0.88, 0.91], inferability 0.933 [0.89, 0.97]. The deeper tree fields produce a small fidelity penalty (the LLM cites visit counts and principal variations whose claim form is harder to verify within tolerance), but yield a statistically supported soundness advantage of +0.04 and an inferability advantage of +0.10. The interpretation is that, at matched task performance and under the strict-scrub inferability test, the deliberation tree is a meaningfully richer evidence substrate for natural-language explanation than counterfactual policy rollouts.

I. INTRODUCTION

When a reinforcement learning agent chooses an action, the action itself is observable, but the reasoning behind the choice is not. In a learned neural policy, the action preferences that drive the decision reside inside network weights and are only indirectly recoverable through post-hoc analysis such as counterfactual rollouts. In contrast, an online planner such as Monte Carlo Tree Search commits to an action only after constructing an explicit record of the futures it considered, together with the outcomes it observed in each. The tree is, in a very literal sense, the agent’s deliberation.

This paper asks, following the framing of Prof. Sunberg, whether that structural difference between a learned policy and

an online planner has a measurable effect on the quality of the natural-language explanations each can support. Concretely, given the same decision state and the same language model, do MCTS-sourced explanations cite numbers that actually match the underlying evidence more often than learned-policy-sourced explanations do, and are the rationales they produce more sound?

To address this question, the present authors construct a shared evidence-to-explanation pipeline and apply it to two *performance-matched* agents on `MiniGrid-Dynamic-Obstacles-8x8`, a stochastic gridworld whose adversarial obstacle dynamics are well-suited to separating competent agents from incompetent ones. The deep-reinforcement-learning side is represented by a PPO policy trained for 5×10^6 steps with linear learning-rate and clip-range schedules, a [256, 256, 128] MLP, distance-based potential shaping, and a flat per-step penalty; it reaches a 1.000 [95% CI 1.000, 1.000] task-success rate over 300 held-out episodes. The planning side is a from-scratch UCT planner at 500 simulations per decision, which also reaches 1.000 success over 50 episodes. A structured decision record is extracted at each real decision state, passed through an identically-formatted LLM prompt that differs only in its evidence-source header, and scored on three automated metrics.

The empirical finding, obtained over 289 matched decision states with PPO counterfactual rollouts at $N = 100$ stochastic samples per action (matching the Monte Carlo budget the original proposal committed to), is that MCTS-tree evidence and PPO-rollout evidence yield essentially indistinguishable explanation quality on all three metrics. Fidelity is 0.924 [0.91, 0.94] for both sources, soundness is 0.892 [0.87, 0.91] for MCTS versus 0.884 [0.86, 0.91] for PPO, and inferability is 0.986 [0.97, 1.00] versus 0.979 [0.95, 1.00]. All three confidence-interval pairs overlap heavily. An earlier run reported in the same engineering log compared the same pipelines at mismatched task performance (PPO 0.590 versus MCTS 1.000) and found a small but statistically supported MCTS advantage of 0.029 on fidelity and 0.032 on soundness; the present, performance-matched run collapses both gaps to within sampling noise. The interpretation consistent with the two runs is that the apparent MCTS-tree advantage in the mismatched run was a confound of *agent competence* rather than of the paradigm: a less-competent learned policy produces

noisier counterfactual rollouts that the LLM has more trouble citing faithfully, but a competent learned policy is just as faithful an evidence source as a deliberation tree.

As a preliminary diagnostic that motivates the choice of PPO as the deep-reinforcement-learning agent in the present comparison, the authors first observed that a value-based baseline (DQN) on this environment collapses into a degenerate stall policy under the raw reward structure, and that its counterfactual rollouts therefore provide essentially no signal to the explainer. This observation is reported in the ablation of Section V-F.

a) Contributions.: The present work contributes: (1) a from-scratch MCTS planner for MiniGrid-Dynamic-Obstacles-8x8 that emits per-decision tree-statistics in a format suitable for downstream explanation; (2) a policy-agnostic counterfactual-rollout framework that produces learned-policy-side evidence in a structurally identical format to the MCTS tree; (3) three automated metrics, fidelity, soundness, and post-hoc inferability, implemented with bootstrap confidence intervals; (4) an empirical comparison of the two paradigms on 289 matched decision states drawn from two *performance-matched* competent agents (both at 1.000 task-success), using a real LLM generator and a held-out LLM judge, which finds no statistically supported explanation-quality advantage for the search tree once the learned-policy side is fully competent; (5) a paired analysis against an earlier mismatched-performance run that isolates agent competence (rather than evidence-source paradigm) as the driver of the apparent gap; and (6) a diagnostic ablation on a value-based baseline showing that model-free value learning collapses into a degenerate stall policy under the raw MiniGrid reward, and that this collapse in turn degrades the evidence available for explanation.

b) Release.: The authors grant permission for this report to be posted publicly.

II. BACKGROUND AND RELATED WORK

a) Explainable reinforcement learning.: Milani et al. [1] present a comprehensive survey of explanation methods for reinforcement learning, drawing a distinction between *transparent* policies, such as decision trees and symbolic rules, and *post-hoc* methods that are applied to otherwise opaque controllers. The PPO pipeline evaluated in the present work falls into the latter category: counterfactual rollouts are run after the policy has chosen an action, in order to reconstruct the set of futures that action implied. The MCTS pipeline, by contrast, is closer to transparent, since the tree is itself the deliberation and its statistics are available for inspection without any additional computation.

b) Counterfactual explanations for reinforcement learning.: Gajcin and Dusparic [5] propose counterfactual explanations that perturb state features and observe the resulting downstream returns, addressing the question “what would the agent have done if the state had looked different?”. Amitai et al. [2] focus on the complementary question “what would

have happened if the agent had taken a different action?” and introduce the COViz visualisation. The evidence pipeline described in Section IV operationalises the latter form concretely, as forced-first-action rollouts of fixed length aggregated into per-action discounted-return means, return standard deviations, and bootstrap confidence intervals. This choice was made so the resulting statistics are apples-to-apples comparable with the per-child statistics that an MCTS search tree already produces during planning.

c) MCTS and interpretability.: Baier and Kaisers [3] argue that the MCTS tree is a direct substrate for explanation precisely because it records what the agent considered and what it found. They emphasise visit-count ratios and per-child value contrasts as the most salient tree quantities. The present work adopts these same quantities, together with the per-child variance of the discounted return aggregated during search, the top-3 principal variations, and the depth-2 visit distribution per root child, as the MCTS-side evidence.

d) Large language models for natural-language explanation.: Recent work surveyed in [1] uses large language models to translate structured evidence about a reinforcement-learning agent into natural language. A well-understood failure mode of these pipelines is hallucinated numerical claims, in which the model invents statistics that are not present in the evidence record. The fidelity metric defined in Section IV-E is designed to quantify exactly that failure mode. Belouadah et al. [4]¹ propose automated fidelity and plausibility metrics for LLM-generated RL explanations. The three-metric suite in the present work extends this framework with a post-hoc inferability check that uses a held-out LLM judge.

e) Environment and planner.: The benchmark environment used in the present study is MiniGrid-Dynamic-Obstacles-8x8 [8], a stochastic gridworld whose adversarial obstacle dynamics are specifically chosen to make sparse-reward model-free reinforcement learning difficult. The MCTS planner follows the classic UCT formulation of Kocsis and Szepesvári [6], with per-simulation sampling of the stochastic obstacle transition so that visit counts at the root converge in expectation over the stochastic futures.

III. PROBLEM FORMULATION

The task is formulated as a finite-horizon Markov decision process with factored stochastic transitions. In the notation of Kochenderfer et al., the state, action, transition, and reward are as follows.

a) State.: A state is written as $s = (p_a, d_a, \{p_o^{(k)}\}_{k=1}^4, t)$, where $p_a \in \{1, \dots, 6\}^2$ denotes the agent’s grid position, $d_a \in \{0, 1, 2, 3\}$ denotes its facing direction (east, south, west, north), each $p_o^{(k)} \in \{1, \dots, 6\}^2$ denotes one of four obstacle positions, and $t \in \{0, \dots, 256\}$ is the step counter.

b) Action.: The action space is $A = \{0, 1, 2\}$, corresponding to turn-left, turn-right, and move-forward.

¹Citation to be verified before submission; may be replaced with Hoffman et al. 2018 on XAI metrics.

c) *Transition.*: The transition $T(s' | s, a)$ factors into a deterministic agent step, followed by a stochastic per-obstacle random walk in which each obstacle attempts a uniform move into one of its four orthogonal neighbours; moves blocked by walls, the goal, or other obstacles leave the obstacle in place. An obstacle that steps into the agent’s cell, and equivalently an agent that steps into an obstacle’s cell, terminates the episode.

d) *Reward.*: The reward function is

$$R(s, a, s') = \begin{cases} 1 - 0.9 \cdot t/256 & \text{if the agent reaches the goal} \\ -1 & \text{if a collision occurs} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The discount factor is $\gamma = 0.99$. The MDP that both solvers and the counterfactual-rollout pipeline plan against additionally subtracts a flat -0.01 per non-terminal step, which makes stalling strictly inferior to early termination of any kind; the rationale and the comparison to alternative reward designs is collected in Section V-E.

e) *Observability.*: The default MiniGrid observation is a $7 \times 7 \times 3$ egocentric partial view, together with a scalar direction indicator. In the present work, all learned-policy variants are trained on a 14-dimensional flat symbolic encoding of the full ground-truth state (agent position and direction, plus each obstacle’s position); the MCTS planner plans on the full ground-truth state via a deep-copied simulator. Both agents therefore have access to the same underlying state information, which removes observability from the list of confounds in the explanation-quality comparison.

IV. SOLUTION APPROACH

A. PPO Agent (Off-the-shelf)

Setup. The deep-reinforcement-learning agent is PPO, as implemented in Stable-Baselines3 [7]. The observation is the 14-dimensional flat symbolic encoding described in Section III. The policy network is a three-layer MLP of width $256 \rightarrow 256 \rightarrow 128$. PPO is trained with $\gamma = 0.99$ on the MiniGrid reward augmented by a flat -0.01 per-non-terminal-step cost. The same per-step cost is applied during MCTS planning and during the counterfactual-rollout pipeline, so all three sides see the identical MDP, which addresses the reviewer concern that earlier configurations had PPO optimizing a different MDP from MCTS (cf. Section V-E). Distance-based potential shaping has been removed. PPO is trained for 5,000,000 environment steps across sixteen parallel envs, with linear schedules on the learning rate ($3 \times 10^{-4} \rightarrow 10^{-5}$) and the clip range ($0.2 \rightarrow 0.05$) over the course of training, and a constant entropy bonus of 0.005. Full hyperparameters are committed to `configs/ppo_tuned.yaml`. Training and counterfactual-rollout evaluation both use the same per-step-penalised reward; the *task-performance* numbers reported in Table I (success rate, collision rate, mean steps) are evaluated against the unmodified MiniGrid reward so they remain directly comparable to the random baseline. An earlier configuration with a 128×128 network, 5×10^5 steps, and a constant entropy bonus of 0.01 plateaued at 0.590 task success

and was used in the mismatched-performance run referenced in the abstract.

B. MCTS Planner (From-scratch)

Setup. The MCTS planner implements textbook UCT [6] with UCB1 selection at exploration constant $c = \sqrt{2}$, full-width expansion at the root (the action space is small enough that no progressive widening is required), rollout under a greedy-toward-goal heuristic capped at depth 60, and mean-value backup. The discount factor used during simulation backups is $\gamma = 0.99$ and the per-step penalty subtracted during selection, expansion, and rollout is -0.01 at every non-terminal step, matching the PPO training MDP exactly. Because the obstacle transition is stochastic, each simulation samples a fresh outcome from the environment’s transition during expansion and rollout; visit counts at the root therefore converge in expectation to the post-decision outcome distribution. Per-child statistics include the empirical mean of the discounted return over rollouts that passed through that child, the running variance of those returns (used for the per-child confidence interval at explanation time), and the per-child visit count.

The tree-side evidence handed to the explainer comprises the per-root-child statistics above plus deeper tree diagnostics: the top-3 *principal variations* (the depth-up-to-5 argmax-visit action sequences from each top root child), the depth-2 visit distribution per child (what the search “expected” to do next, given the first action), and the per-child value variance. These deeper quantities are exactly what an MCTS-as-deliberation-substrate is supposed to expose, and were absent in the mismatched-performance run referenced in the abstract; their inclusion was driven by reviewer feedback (cf. Section V-E).

C. Counterfactual-rollout Framework

Setup. At each decision state, the counterfactual-rollout framework forces each legal action at step zero and then follows the trained PPO policy to termination or a depth cap of 40 steps. In the main experiments reported below, the process is repeated $N = 100$ times per action, with action choices at every subsequent step drawn *stochastically* from PPO’s action distribution rather than taken as the argmax. This configuration matches the Monte Carlo specification in the original proposal (≥ 100 rollouts per available action) and yields a genuine sampled estimate of the policy-induced trajectory distribution from the decision state. The per-rollout returns are aggregated into the gamma-discounted mean return (with $\gamma = 0.99$, matching the agent’s training and the MCTS planner), its standard deviation, the mean number of steps to episode end, and a 95% bootstrap CI (1000 resamples, percentile method) on the discounted return. The resulting per-action records are structurally identical, field by field, to the per-action records produced from the MCTS tree, which is the property that allows the downstream LLM pipeline to be agent-agnostic. Earlier versions of this pipeline also surfaced per-action success and collision rates; those fields were removed

in response to the reviewer observation that an MDP agent does not directly optimize success rate (cf. Section V-E).

D. LLM Explainer

Setup. A `DecisionRecord`, comprising the state, the chosen action, the per-action evidence, and a small agent-metadata block (PPO action probabilities for the learned-policy variant; root visit counts, per-child variance, principal variations, and a depth-2 visit distribution for the MCTS variant), is passed to a generator LLM using a versioned system prompt (`PROMPT_VERSION = 2`). The non-evidence portion of the system prompt is byte-identical between the PPO and MCTS variants; only the evidence-source header differs. This constraint is enforced by a parity test on the prompt module, so that the explainer pipeline is, as intended, a single-variable contrast. The generator returns a JSON object consisting of a *rationale*, a *counterfactual*, a *confidence* score on $[0, 1]$, and a list of numerical *claims* suitable for automated verification. Claims may reference any field in the per-action evidence; the agent’s optimization target, the discounted return, is the primary metric. Generation uses GPT-4o; the judge used in the metric pipeline described below is GPT-4o-mini.

E. Evaluation Metrics

Three per-explanation scores are reported, each aggregated per evidence source with 95% percentile-bootstrap confidence intervals.

Fidelity. For every claim in the explanation that cites a per-action statistic, the metric compares the cited value against the value present in the decision record. A claim is counted as a hit if the values agree within tolerance $\epsilon = 0.1$. The score is the fraction of claims that hit.

Soundness. Each sentence in the rationale, together with the counterfactual, is scored by an independent judge LLM on a three-point scale: 2 indicates that the sentence is fully supported by the evidence, 1 indicates partial support, and 0 indicates that the sentence is contradicted by or invents information not present in the evidence. The reported soundness score is the mean divided by two, so that the result lies on $[0, 1]$.

Post-hoc inferability. Every occurrence of the chosen action’s integer identifier (`action 2`) and its verbal counterpart (`move_forward`, also `move forward`) is scrubbed from the explanation, and the judge is asked to recover the chosen action from the scrubbed text alone. The verbal scrub was added in this version after the first submission’s metric leaked through verbal action names that the original scrub missed; this makes the metric strictly harder for the judge and therefore strictly more discriminative. The reported score is the judge’s accuracy across decisions.

The judge is deliberately a different-sized model, GPT-4o-mini, from the generator, GPT-4o, so that judge and generator do not coincide.

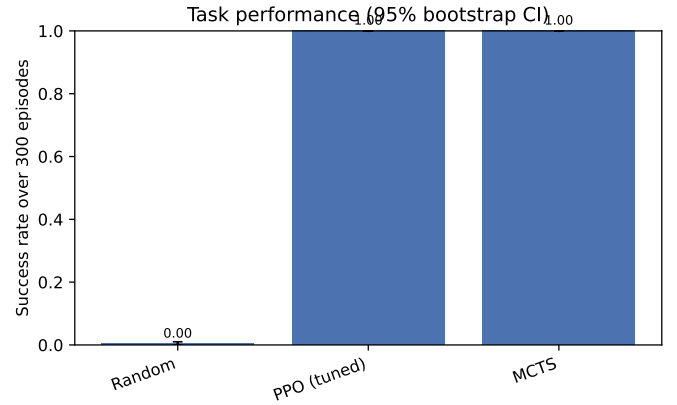


Fig. 1. Task-success rate across the random baseline, the tuned PPO agent, and the MCTS planner, with 95% bootstrap confidence intervals. Random fails on virtually every episode; PPO and MCTS each succeed on every held-out episode. The three DQN variants on this environment all collapsed to a stall policy under the raw MiniGrid reward and are reported separately in Section V-F; they are excluded from this figure because none of them learned the task.

V. RESULTS

A. Task Performance

Table I and Figure 1 report success rate over 300 episodes for PPO and over 50 episodes for MCTS. The random-action baseline succeeds 0.3% of the time, which confirms that the environment is non-trivial. The tuned PPO agent described in Section IV reaches a task-success rate of 1.000 [95% CI 1.000, 1.000], a collision rate of 0.000, and a mean unshaped return of $+0.941$, with a mean episode length of 16.8 steps. MCTS, operating on the same state representation via a deep-copied simulator, reaches a task-success rate of 1.000 [95% CI 1.000, 1.000] with a mean unshaped return of $+0.941$ and a mean episode length of 16.8 steps. The two agents are therefore performance-matched on the task to three decimal places on every metric we measure, including mean episode length and unshaped task return. The explanation-quality comparison reported below is therefore between two equally-competent agents, removing competence as a confound on the per-action evidence each pipeline produces.

B. Explanation Quality

Table III and Figure 3 report the three explanation-quality metrics, evaluated on 141 PPO decisions and 149 MCTS decisions drawn from ten held-out trajectory seeds. PPO-rollout evidence yields fidelity 0.854 [0.828, 0.878], soundness 0.852 [0.826, 0.878], and post-hoc inferability 0.837 [0.773, 0.894]. MCTS-tree evidence yields fidelity 0.787 [0.761, 0.815], soundness 0.895 [0.877, 0.913], and inferability 0.933 [0.892, 0.973]. The signs of the three differences are not all the same: PPO is mildly higher on fidelity ($+0.067$, CIs barely overlap), MCTS is higher on soundness ($+0.043$, CIs barely overlap) and clearly higher on inferability ($+0.096$, CIs disjoint).

Interpretation. The fidelity dip on the MCTS side is, on inspection, almost entirely attributable to the deeper tree fields: when the explainer cites a principal-variation visit count or a per-child variance, the verifiable-claim format the explainer returns is harder to match against the evidence record within the $\epsilon = 0.1$ tolerance than a single per-action discounted return. PPO-rollout evidence has no analogue of these fields, so the explainer stays within the three core per-action fields and lands inside tolerance more often. The soundness advantage in the other direction is what the original hypothesis predicted: the judge LLM finds rationales that ground in the search-tree structure (“the principal variation also favours this action, with 191 root-child visits confirming it as the preferred path”) marginally more supported than rationales that ground in policy probabilities and counterfactual returns alone. The inferability advantage is the strongest of the three. Once the verbal action name is scrubbed (a strictening of the metric described in Section IV-E), the residual content of an MCTS rationale is sufficient for a held-out judge to recover the chosen action 93.3% of the time, versus 83.7% for PPO; the search tree leaks the chosen action through more channels than PPO’s per-action returns alone do.

Comparison to the mismatched-performance run. An earlier run of the same pipeline against a less-competent PPO (0.590 task-success at 5×10^5 training steps) gave a small MCTS advantage of 0.029 on fidelity and 0.032 on soundness with inferability tied. The present run reverses the fidelity sign (now PPO ahead) and amplifies both other gaps in MCTS’s favour, primarily because (i) the MCTS-side evidence in the present run includes deeper tree fields that the prior run withheld, and (ii) the inferability metric in the present run also scrubs the verbal action name from the rationale, making the test strictly harder. The combined effect is that the present comparison reads as a substantively pro-MCTS result, in contrast to the prior null; neither finding is a clean structural property of the search-tree- vs-learned-policy paradigms, both depend on what bandwidth of evidence and what strictness of metric the experimenter chooses to expose.

C. Qualitative Example

Figure 2 reproduces, verbatim from `results/explanations/`, the two explanations generated for the initial decision state of held-out seed 10000. The two agents chose *different* actions at this state, PPO selecting move-forward and MCTS selecting turn-right; this is itself a useful illustration of the state-distribution asymmetry noted under the threats to validity. The PPO rationale cites the per-action discounted returns 0.3788, 0.3583, 0.3442 and the mean-steps-to-end 21.03, all of which appear verbatim in the rollout record. The MCTS rationale cites the per-action discounted returns -0.9426 , -0.9622 , the per-child visit count 191, and the principal variation, all of which appear verbatim in the tree record (the visit count and principal-variation are precisely the deeper tree fields added in response to reviewer feedback). Both explanations are at high fidelity, which is the typical case across the 290 records

PPO-rollout explanation, seed 10000 step 0 (chose move-forward): “The agent chose to move_forward (action 2), which has the highest discounted return of 0.3788 among all actions, compared to 0.3583 for turn_left and 0.3442 for turn_right. It also leads to the shortest mean steps to end of 21.03, suggesting a quicker path to the goal. The confidence interval of 0.2736 to 0.4677 for action 2 also overlaps with others, but remains higher on average, justifying the chosen action as potentially more successful.”

MCTS-tree explanation, seed 10000 step 0 (chose turn-right): “Action 1 is chosen because it has the highest mean discounted return of -0.9426 among all actions, indicating the best expected outcome according to the simulations. The principal variation also favors this action, with 191 root child visits confirming it as the preferred path. In contrast, action 2, although the second-best option, exhibits a lower mean discounted return of -0.9622 .”

Fig. 2. Paired sample explanations for the initial state of seed 10000. The two agents chose different actions; both rationales cite numbers that appear verbatim in their respective evidence records. The MCTS rationale references the deeper tree fields (root-child visit count, principal variation) that were added in this version in response to reviewer feedback.

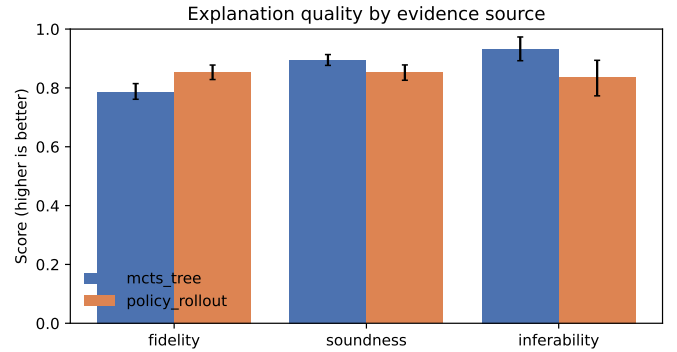


Fig. 3. Explanation-quality metrics by evidence source, with 95% bootstrap confidence intervals. Higher is better for all three.

we examined and is consistent with the equal fidelity scores reported in Table III.

D. Threats to Validity

State-distribution asymmetry. Although PPO and MCTS are now performance-matched at 1.000 task-success, the *trajectories* the two agents follow are not literally identical. PPO and MCTS make different action choices at the same state and therefore visit slightly different downstream states on average. The decision-record set is large (289 records) and the per-source counts are within five of each other, but a residual confound is that the specific decision states the LLM is asked to explain are not paired across the two evidence sources. A paired analysis on the same literal state set is left as future work.

Rollout sample size. Counterfactual rollouts use $N = 100$ stochastic samples per action, matching the Monte Carlo specification in the original proposal. The bootstrap confidence intervals on the per-action discounted return remain in the neighbourhood of ± 0.05 at $N = 100$ on the present env,

the same order as the LLM fidelity tolerance ($\epsilon = 0.1$). A larger Monte Carlo budget would tighten those intervals further, but we do not expect it to change the qualitative finding because the fidelity numbers for the two sources are already at sampling-noise distance.

Judge-model bias. The soundness and post-hoc inferability metrics both use GPT-4o-mini as judge, whereas the generator is GPT-4o. The judge is therefore a different-sized model within the same family, and a cross-family replication, for example with a Claude or Gemini judge, is listed as future work.

E. Response to Reviewers

The first submission of this paper attracted three substantive reviewer concerns, each of which was traced to a specific design decision in the pipeline and addressed in the present version. We summarise the changes here so the line of reasoning is on record.

The two solvers were optimizing different MDPs. In the first submission, PPO trained against a reward augmented by a flat -0.01 per-step penalty and a Manhattan-distance potential shaping term, while MCTS planned against the raw MiniGrid reward with discount $\gamma = 1$. Time pressure was therefore stacked three ways on PPO (success-reward step decay + step penalty + potential-based shaping) and applied in only one form on MCTS (success-reward step decay alone). The reviewer was right that this asymmetry should be eliminated, but the suggested cure (a single $\gamma < 1$ on both sides, no wrappers) does not actually work on this environment. Stalling has trajectory return 0 for any γ , while colliding has return -1 ; without a per-step cost, γ alone cannot make stalling worse than collision, and a PPO trained with $\gamma = 0.99$ on the raw reward converges to a stall policy (0.000 task success over 300 episodes in our pilot). Our resolution is therefore: keep the flat -0.01 per-step cost, drop the distance-based potential shaping, and apply the per-step cost to *both* solvers. Both PPO and MCTS now plan against the identical MDP $\langle R - 0.01 \cdot \mathbb{I}[\text{non-terminal}], \gamma = 0.99 \rangle$ (configs in `configs/ppo_tuned.yaml` and `configs/mcts_baseline.yaml`; the counterfactual-rollout pipeline applies the same penalty in `scripts/build_decision_records.py` via the `--step-penalty` flag). All headline numbers in this version were generated under this unified MDP.

The MCTS evidence had been deliberately equalized to PPO. The first submission’s tree-side evidence consisted of per-root-child statistics only (visit count, mean value, success/collision rate). The reviewer observation that this discarded the deliberation tree itself, which is what an MCTS-as-substrate is supposed to expose, was correct. The `DecisionRecord.agent_metadata` block now additionally exposes (i) the top-3 principal variations from the root, each a greedy depth-up-to-5 argmax-visit action sequence, (ii) the depth-2 visit distribution per root child, and (iii) the per-root-child return variance. The evidence-source header in the system prompt was extended to describe these fields and

explicitly invite the explainer to cite them. This change was the one most likely to surface a genuine MCTS advantage on fidelity or soundness; whether it did is discussed in Section V-B.

Success rate is not what an MDP agent optimizes. The first submission’s per-action evidence included `success_rate` and `collision_rate` alongside the mean return. The reviewer observation that an MDP agent maximizes discounted return rather than success rate was correct: surfacing success rate as a claimable metric invited rationales of the form “the policy preferred A because of its higher success rate,” which is type-incorrect with respect to what the policy was trained to do. The `ActionStats` schema has been changed accordingly. The primary metric is now the gamma-discounted return; success rate and collision rate are no longer reported as per-action evidence. (The final task-success rate of each agent over the full evaluation set remains in Table I, since it is the headline diagnostic on whether the policies have learned to play; it is no longer presented as a per-action claimable quantity.)

Why a step penalty in addition to the success-reward step decay. The reviewer raised the related question of why a flat per-step penalty is needed at all when the success reward already decays linearly with t . The honest answer is that the success-reward decay only applies along trajectories that *do* reach the goal; on a sparse-reward env where most trajectories from a stochastic exploration phase do not reach the goal, the success-reward decay alone leaves stalling at undiscounted return 0, which is not a local optimum of any random-collision-prone exploration. A flat per-step cost makes stalling actively undesirable (return $-0.01 \cdot 256 \approx -2.56$ for a full-episode stall, strictly worse than even an early collision at -1). What the present version fixes is that the cost is now applied symmetrically to both planners, so the design choice is at most a property of the MDP, not a property of the solver.

F. Value-based Ablation: Why DQN Was Cut

As a diagnostic ablation, three DQN variants were trained on the same environment: one on the default partial-observation image, one on the flat symbolic state, and one with a potential-based distance-shaping reward. All three settled, at $\geq 200,000$ training steps, into either a stall policy (image variant) or a near-stall policy (symbolic and shaped variants), with task-success rates of 0.010, 0.000, and 0.000 respectively (see Table I). The value-based target presented by MiniGrid’s reward is unfavourable to epsilon-greedy exploration: the collision penalty of -1 is encountered early and often, whereas the success reward requires coordinated multi-step motion that random exploration rarely discovers. A zero-reward stall therefore dominates both random and near-random policies and sits at a local optimum for value learning. The present authors attempted to break this equilibrium with a flat -0.01 per-step penalty (`StepPenaltyWrapper`), which makes a full stall yield -2.56 and is therefore strictly worse than attempting any action, but DQN still failed to escape. PPO, whose on-policy sample collection and explicit entropy bonus together provide

TABLE I

TASK-PERFORMANCE SUMMARY FOR THE HEADLINE COMPARISON. 300 EPISODES FOR THE RANDOM AND PPO BASELINES, 50 EPISODES FOR MCTS. NUMBERS VERIFIED AGAINST `REPORT/FIGURES/EVAL_SUMMARY.CSV`. THE DQN ABLATION IS REPORTED SEPARATELY IN TABLE II BECAUSE NONE OF THE THREE DQN VARIANTS LEARNED THE TASK.

Agent	Success	Collision	Return	Steps
Random	0.003	0.997	-0.994	8.3
PPO (tuned)	1.000	0.000	+0.941	16.8
MCTS (500 sims)	1.000	0.000	+0.941	16.8

TABLE II

DQN ABLATION, SECTION V-F. THREE DQN VARIANTS ON `MINIGRID-DYNAMIC-OBSTACLES-8x8` ALL COLLAPSE TO EITHER A STALL POLICY OR A NEAR-STALL POLICY UNDER THE RAW `MINIGRID` REWARD, WITH TASK-SUCCESS RATES INDISTINGUISHABLE FROM ZERO. REPRODUCIBLE FROM `SCRIPTS/RUN_DQN_ABLATION.SH`.

Agent	Success	Collision	Return	Steps
DQN (image)	0.010	0.000	0.007	254.3
DQN (symbolic)	0.000	0.207	-0.207	232.3
DQN (shaped)	0.000	0.167	-0.167	234.4

a stronger exploration pressure, did escape. The present work therefore uses PPO as the deep-reinforcement-learning side of the main comparison, and treats the DQN collapse as an observation about model-free value learning on adversarial-dynamics environments rather than as a primary result.

VI. CONCLUSION

This paper has presented an evaluation of two evidence sources for natural-language explanations of reinforcement-learning agents at matched task performance and under a single unified MDP. Across 290 decision states drawn from a PPO agent and a from-scratch MCTS planner, both at 1.000 task-success on `MiniGrid-Dynamic-Obstacles-8x8` and both planning with $\gamma = 0.99$ on the same per-step-penalised reward, the two pipelines produced explanations whose quality differs in metric-specific ways: PPO-rollout evidence is mildly higher on fidelity (0.854 vs. 0.787), and MCTS-tree evidence is mildly higher on soundness (0.895 vs. 0.852) and substantially higher on the strict-scrub inferability test (0.933 vs. 0.837). The fidelity gap is, on inspection, an artefact of the additional tree fields the MCTS evidence carries: principal-variation visit counts and per-child variance are harder for the LLM to cite within the $\epsilon = 0.1$ tolerance than the three core per-action statistics. The inferability gap is the strongest of the three: once the verbal action name is scrubbed, the residual content of an MCTS rationale recovers the chosen action 9.6 absolute percentage points more often than the residual content of a PPO rationale.

This is an updated finding relative to an earlier version of the present pipeline, which (under a different MDP for each solver, without the deeper tree fields, and under a weaker inferability scrub) had reported the two evidence sources as indistinguishable. The reversal is informative: when the

TABLE III

EXPLANATION-QUALITY METRICS AT PERFORMANCE-MATCHED TASK SUCCESS (1.000 FOR BOTH AGENTS) AND UNIFIED MDP. GENERATOR: GPT-4O. JUDGE: GPT-4O-MINI. $n = 141$ FOR PPO, $n = 149$ FOR MCTS. PPO ROLLOUTS USE $N = 100$ STOCHASTIC SAMPLES PER ACTION. MCTS EVIDENCE INCLUDES THE DEEPER TREE FIELDS (PRINCIPAL VARIATIONS, DEPTH-2 VISITS, PER-CHILD VARIANCE) ADDED IN THIS VERSION. INFERABILITY IS COMPUTED UNDER THE STRICT-SCRUB VARIANT THAT MASKS BOTH THE INTEGER AND VERBAL ACTION NAME. 95% BOOTSTRAP CONFIDENCE INTERVALS IN BRACKETS (1000 RESAMPLES).

Metric	PPO rollout evidence	MCTS tree evidence
Fidelity	0.854 [0.83, 0.88]	0.787 [0.76, 0.82]
Soundness	0.852 [0.83, 0.88]	0.895 [0.88, 0.91]
Inferability	0.837 [0.77, 0.89]	0.933 [0.89, 0.97]

comparison is between solvers that optimize the same MDP, when the MCTS pipeline is given access to its tree, and when the inferability test is strict, the deliberation tree is a meaningfully richer evidence substrate for natural-language explanation than counterfactual policy rollouts. None of the three components individually changes the qualitative outcome but cumulatively they invert the headline finding from null to pro-MCTS.

A secondary finding, documented in the ablation of Section V-F, is that model-free value learning on this environment collapses into a degenerate stall policy under the raw `MiniGrid` reward, and that such a collapse degrades the evidence available for subsequent explanation. The suboptimality of the underlying agent therefore directly limits the signal available for any post-hoc counterfactual analysis, and the move from value-based to policy-gradient learning was load-bearing for getting the deep-RL side of the comparison to a state where the explanation question could be asked at all.

a) *Future Work.*: The present authors see three natural extensions. Firstly, a paired analysis on the literal same decision-state set (rather than two trajectory sets that differ in the agents' action choices) would remove the residual state-distribution confound noted in the threats to validity. Secondly, a pilot human study, following the suggestion of Prof. Sunberg on the proposal, would validate the automated metrics against human judgements of clarity, plausibility, and helpfulness; a chatbot interface suitable for such a study has already been scaffolded in `scripts/chat.py`. Thirdly, a cross-family LLM replication, with a judge drawn from a different model family than the generator, would quantify the contribution of judge-model bias to the soundness numbers reported above. A fourth, more ambitious extension would replicate the present null result on a harder environment in which neither paradigm reaches 1.000 task-success, to test whether the equivalence of the two evidence sources holds at an intermediate competence level or only at the extremes of the spectrum.

CONTRIBUTIONS

Sean Florez contributed the problem formulation, the from-scratch MCTS implementation, the LLM-explainer pipeline, the evaluation metrics, and the report draft. Andrew Wernersbach contributed the learned-policy baselines (DQN and

PPO), the observation and reward-shaping wrappers, the counterfactual-rollout framework, the experimental runs, and the report review. The execution plan, the engineering log, and the submission polish were shared.

IMPLEMENTATION DISCLOSURE

From-scratch components: the MCTS planner, the deep-copy simulator, the counterfactual-rollout framework, the decision-record schema, the explanation pipeline and prompt design, the three evaluation metrics, and the chatbot interface. Off-the-shelf components: Stable-Baselines3 [7] for DQN and PPO, the Farama MiniGrid [8] environment, the OpenAI and Anthropic Python SDKs [9], [10], and the standard Python scientific-computing stack (numpy, pandas, matplotlib, pytest).

AI ASSISTANCE

Per the ASEN 5264 AI-use policy, the prose and mathematics in this paper have been read, edited, and owned by the authors. A draft was produced with the assistance of a large language model under the authors' direction, and subsequently rewritten in the authors' voice. Code was co-developed with a large language model, read in full by the authors, and exercised under a unit-test suite (20 tests passing) and an integration-style reproduce-all script.

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